

2 MARKS

1) **Expand and define the term LASER?**

Ans: L - light
A - Amplification
S - Stimulated
E - Emission
R – Radiation.

Laser is a device which emits monochromatic highly coherent beam of light.

2) **Define spontaneous emission?**

Ans: The process in which an atom moves from excitation state to ground state after completion of its life time period that is 10^{-8} sec.

3) **Define optical fiber?**

Ans: Optical fiber is a thin fiber made of thin dielectric material like plastic or glass through, which light signal is transmitted with very little loss of strength / data / information.

4) **Define nanoparticles?**

Ans: Nanoparticles are tiny particles with dimensions ranging from 1 to 100 nanometers. They exhibit unique properties due to their small size and large surface area to volume ratio.

5) **Expand SEM, TEM, AFM, and FTIR?**

Ans: Scanning Electron Microscope (SEM)
Transmission Electron Microscope (TEM)
Atomic Force Microscope (AFM)
Fourier Transform Infrared Spectroscopy (FTIR)

6) **Define Heat?**

Ans: Heat is a form of energy that transfers from one substance to another substance.

7) **Define crystal?**

Ans: A crystal is a solid material where the constituent atoms, molecules, or ions are arranged in a highly ordered, repeating microscopic structure, forming a crystal.

8) **State Newton's second law of motion?**

Ans: The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force.

9) **Write any four properties of laser?**

Ans: 1 Monochromatic in nature.
2 Sharp in nature.
3 Travel a long distance.
4 Highly directional in nature.
5 Highly coherent in nature.

10) **Define Thermodynamic?**

Ans: Thermodynamic is the branch of science that studies the relationships between heat, temperature and energy in physical and chemical systems.

5 MARKS

1) Explain the construction and working of He-Ne laser with the neat diagram?

Ans: Gas Laser: He-Ne Laser

Diagram

Construction:

1. A He-Ne laser consists of a discharge tube.
2. The length of the discharge tube is 80 cm and the diameter is 1 cm.
3. The discharge tube is made up of quartz material.
4. The ratio of He: Ne gas is 10: 1 ($\text{He} > \text{Ne}$).
5. The pressure of helium gas is 0.1 mm of Hg and the pressure of neon gas is 1 mm of Hg.
6. Two mirrors are connected to the discharge tube; one is a partially silvered mirror and the other is a fully silvered mirror.
7. Two electrodes are connected to the discharge tube.
8. The output power of the laser depends upon the length of the discharge tube and the pressure of the gas.

Working of He-Ne Laser:

Diagram:

- When a certain amount of voltage is applied to the discharge tube with the help of a **DC power supply**, electrons gain energy.
- The energetic electrons, after entering the discharge tube, collide with helium (He) atoms and transfer their energy to them. These helium atoms are initially in the **ground state**.
- After gaining energy, the helium atoms get excited and move from the **ground state to the excited state**.
- The excitation energy levels of helium and neon are nearly the same. Hence, the excited helium atoms transfer their energy to neon (Ne) atoms.
- As a result, neon atoms get excited to higher energy levels (E_6). Then, neon atoms undergo transitions such as:

Physics Q-Bank

- $E_6 \rightarrow E_5$
- $E_6 \rightarrow E_4$
- $E_6 \rightarrow E_3$
- $E_6 \rightarrow E_2$
- $E_6 \rightarrow E_1$
- During the transition from $E_6 \rightarrow E_4$, a large amount of energy is released in the form of light with a wavelength of **6328 Å**.
- Finally, the emitted photons are reflected back and forth between the **fully silvered mirror** and the **partially silvered mirror**, resulting in amplification.
- The laser beam emerges from the **partially silvered mirror** as the output.

2) Explain the properties of carbon nanotubes?

Ans: **Properties of CNTs (Carbon Nanotubes):**

1. Mechanical Properties:

- The tensile strength of CNTs is greater than that of steel.
- They are approximately **100 times stronger than steel**.
- They are **highly elastic** in nature.

2. Thermal Properties:

- The thermal conductivity of CNTs is **higher than that of diamond**.
- They can retain their physical properties up to **2800°C in vacuum**.
- They are very **stable materials for heat conduction**.

3. Electrical Properties:

- The electrical conductivity of CNTs is **about 100 times greater than copper**.
- They can behave either as **metals or semiconductors**.
- Their properties depend on the **size and structure of the graphite layer**.

3) Explain the types of electromagnetic materials? (From ChatGPT)

Ans: Electromagnetic materials are classified based on how they respond to an external magnetic field. The main types are:

1. Diamagnetic Materials

- These materials are weakly repelled by a magnetic field.
- They do not retain magnetism after removal of the external field.
- Magnetic susceptibility is small and negative.
- **Examples:** Copper, Bismuth, Silver

2. Paramagnetic Materials

- These are weakly attracted by a magnetic field.
- Magnetization exists only in the presence of the external field.
- Magnetic susceptibility is small and positive.
- **Examples:** Aluminium, Platinum, Oxygen

3. Ferromagnetic Materials

Physics Q-Bank

- Strongly attracted by magnetic fields.
- Can retain magnetism even after removal of the external field (permanent magnets).
- Have large positive susceptibility.
- **Examples:** Iron, Nickel, Cobalt

4. Antiferromagnetic Materials

- Adjacent atoms have magnetic moments aligned in opposite directions, cancelling each other.
- Net magnetization is nearly zero.
- **Examples:** Manganese oxide, Nickel oxide

5. Ferrimagnetic Materials

- Similar to antiferromagnetic, but unequal opposite moments result in some net magnetization.
- Widely used in transformers and inductors.
- **Examples:** Ferrites, Magnetite

Conclusion:

These materials differ in their magnetic behaviour and are selected based on applications like electrical machines, sensors, and magnetic storage devices.

4) Write a short note on thermal sensors? (From ChatGPT)

Ans: A thermal sensor is a device used to measure temperature or detect changes in heat energy and convert them into an electrical signal. It plays an important role in engineering systems, electronics, and industrial processes.

Thermal sensors work based on the principle that physical properties of materials—such as resistance, voltage, or expansion—change with temperature. These changes are then measured and converted into readable outputs.

Common types of thermal sensors include:

- Thermocouples – generate voltage due to temperature difference between two junctions.
- Thermistors – semiconductor devices with high sensitivity to temperature changes.

Thermal sensors are widely used in applications such as:

- Automobiles (engine temperature monitoring)
- Home appliances like air conditioners and refrigerators
- Medical devices and environmental monitoring

In conclusion, thermal sensors are essential components for accurate temperature measurement and control, ensuring safety, efficiency, and reliability in various systems.

5) Write a short note on radiation sensors? (From ChatGPT)

Ans: A radiation sensor is a device used to detect and measure ionizing radiation such as alpha (α), beta (β), and gamma (γ) rays. These sensors convert radiation energy into an electrical signal that can be measured and analysed.

Working-Principle:

Radiation sensors operate based on the ionization effect. When radiation passes through a medium (gas, solid, or semiconductor), it ionizes atoms and produces charged particles. These charges are collected to generate an electrical signal proportional to the radiation intensity.

Applications:

Physics Q-Bank

- Medical imaging and cancer treatment
- Nuclear power plants monitoring
- Environmental radiation measurement
- Industrial inspection and safety

Advantages:

- High sensitivity to radiation
- Quick detection and response

Disadvantages:

- Expensive (especially semiconductor types)
- Requires careful handling and calibration

Conclusion:

Radiation sensors are essential devices for detecting harmful radiation and ensuring safety in medical, industrial, and nuclear environments.

6) Explain the construction and working of SEM with neat diagram?

Ans: **Diagram**

Construction:

The following parameters are used:

1. Vacuum chamber
2. Electron gun
3. Anode
4. Condenser Lense
5. Scanning coils
6. Objective lens
7. Sample
8. Sample holder
9. Photo detector
10. P.C.

Working:

Physics Q-Bank

The amount of power supply enters into the vacuum chamber and electron gun made up of tungsten start liberating the electrons. To accelerate the speed of electron beam, we use anode. Those accelerated beams pass through the condenser lens in a particular width and direction. Then those beams are scanned by the scanning coils in the raster form. The scanned beams are passed from objective lens and falls on the sample. The interaction between electron beam and sample takes place which results in emission of secondary electrons. These are collected by a device known as photodetector. Finally, photodetector transfers these signals to computer system by result of which it gives particle size, chemical composition and morphological structures.

7) Explain the different types of thermodynamic systems?

Ans: Types of Thermodynamic System

1) Open System:

The system in which the transfer of mass as well as energy can take place across its boundary is called as open system.

Ex: Turbine, Pump, Compressor.

2) Closed System:

Transfer of energy takes place across its boundary with the surrounding but no transfer of mass takes place.

Ex: Lunch box, Piston cylinder.

3) Isolated System:

Neither the transfer of mass nor that of energy takes place across its boundary with the surrounding.

Ex: Thermos flask.

8) Explain the working of the atomic force microscope (AFM)? (For both 5 & 10 marks)

Ans: Atomic Force Microscope (AFM).

Diagram

Construction:

Physics Q-Bank

1. Cantilever: Made up of metal sheet.
2. Cantilever Tip: is about 7-8 atoms.
3. Sample
4. Sample holder
5. Piezoelectric tube: Moves in all 3 directions.
6. Feedback Controller: Passes the signal to the computer.
7. Laser
8. Photo detector: Used to detect the signals coming from cantilever.
9. PC for display.

Modes of Working

Contact Mode: Interaction between the tip and the sample, for metal sample.

Non-Contact Mode: It does not touch the sample but scans it. For biological samples.

Tapping Mode: Up down interaction b/w tip and the sample. For rough surface samples.

Advantages

It can give 3D image.

Has good resolution.

Does not require vacuum chamber.

Non-conducting samples are also used.

Disadvantages

It takes more time to scan mode.

Distortion of sample in contact mode.

10 Marks

1) Derivation of Einstein's Coefficients

Ans: At thermal equilibrium, the number of upward transitions is equal to the number of downward transitions.

$$B_{12}N_1\rho(\nu) = A_{21}N_2 + B_{21}N_2\rho(\nu)$$

Rearranging

$$\rho(\nu) = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2}$$

Dividing numerator and denominator by $B_{21}N_2$:

$$\rho(\nu) = \frac{\frac{A_{21}}{B_{21}}}{\frac{B_{12}N_1}{B_{21}N_2} - 1} \quad (1)$$

Using Boltzmann Distribution Law:

$$N_i = N_0 e^{-E_i/kT}$$

So,

$$N_1 = N_0 e^{-E_1/kT}, N_2 = N_0 e^{-E_2/kT}$$

$$\frac{N_1}{N_2} = e^{(E_2 - E_1)/kT}$$

Since $E_2 - E_1 = h\nu$,

$$\frac{N_1}{N_2} = e^{h\nu/kT} \quad (2)$$

Substituting (2) into (1):

$$\rho(\nu) = \frac{\frac{A_{21}}{B_{21}}}{\frac{B_{12}}{B_{21}} e^{h\nu/kT} - 1} \quad (3)$$

According to Planck's Radiation Law:

$$\rho(\nu) = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{h\nu/kT} - 1} \quad (4)$$

Comparing equations (3) and (4), we get:

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$$

$$\frac{B_{12}}{B_{21}} = 1$$

Conclusion:

To obtain a good laser source, stimulated emission must be greater than spontaneous emission.

Physics Q-Bank

2) Define thermodynamic process and explain different types of thermodynamic processes?

Ans: Thermodynamics Processes

Thermodynamics process: Change in physical quantities of a system is called thermodynamic process.

Isothermal Process: A process in which the temperature of a system remains constant is called isothermal process.

Isobaric Process: A process in which the pressure of a system remains constant is known as isobaric process.

Isochoric Process: A process in which the volume of the system remains constant is known as isochoric process.

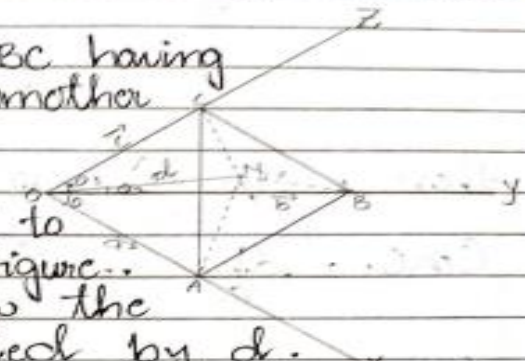
Adiabatic Process: In this process heat will not flow inside the system. Such a process is called adiabatic process.

Microscopic Approach: The behaviour of individual molecule is taken into consideration and this approach is known as statistical thermodynamic approach.

Macroscopic Approach: The behaviour of individual particle/molecule is not taken into consideration but the average behaviour is taken into consideration. This approach is known as classical thermodynamic approach.

Interplanar Spacing In Terms Of Miller Indices

Consider a plane ABC having Miller Indices (h, k, l) & another plane which is passing through origin. Draw a normal ON from point 'O' to the plane ABC as is in figure. ON gives the distance b/w the plane which is represented by d .



From figure,

$$\triangle ONA, \cos \theta_1 = \frac{ON}{OA} \quad \text{--- (1)}$$

$$\triangle ONB, \cos \theta_2 = \frac{ON}{OB} \quad \text{--- (2)}$$

$$\triangle ONC, \cos \theta_3 = \frac{ON}{OC} \quad \text{--- (3)}$$

$$OA, OB, OC$$

$$\frac{OA}{a}, \frac{OB}{b}, \frac{OC}{c}$$

$$\frac{a}{OA}, \frac{b}{OB}, \frac{c}{OC}$$

$$h, k, l$$

$$h = \frac{a}{OA}, \quad k = \frac{b}{OB}, \quad l = \frac{c}{OC}$$

$$OA = \frac{a}{h}, \quad OB = \frac{b}{k}, \quad OC = \frac{c}{l}$$

$$ON = d$$

then, eqn (1), (2) and (3) becomes

$$\cos \theta_1 = \frac{dh}{a}, \quad \cos \theta_2 = \frac{dk}{b}, \quad \cos \theta_3 = \frac{dl}{c}$$

According to Cartesian Co-ordination System.

$$\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3 = 1$$

$$\left(\frac{dh}{a}\right)^2 + \left(\frac{dk}{b}\right)^2 + \left(\frac{dl}{c}\right)^2 = 1$$

$$d^2 \left[\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right] = 1$$

$$d^2 = \frac{1}{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}$$

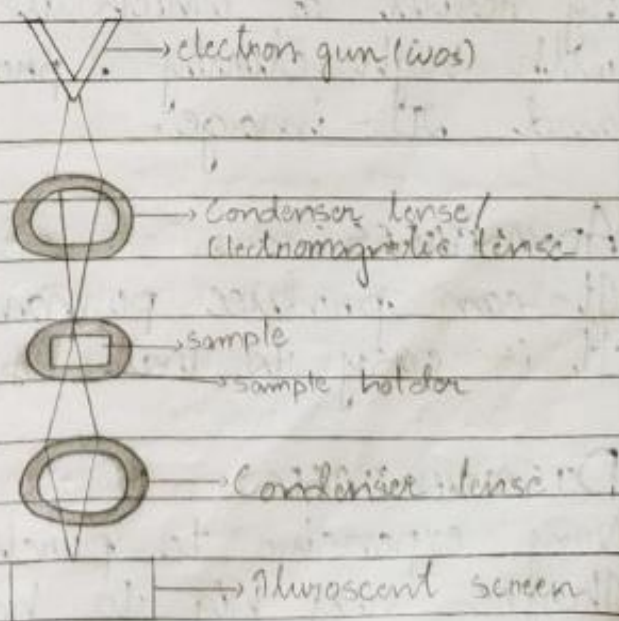
$$= \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$

For Cubic Crystal System:

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Where a is lattice constant.

TRANSMISSION ELECTRON MICROSCOPE (TEM):



CONSTRUCTION :

- Electron gun : Made up of Tungsten
- Condenser lense : Made up of Copper metal, which is also known as Electromagnetic material.
- Sample : Conducting material.
- Sample holder
- Condenser lense : Amplify the electrical signal
- Fluorescent Screen : Converts electrical signal into light signal (Visible light)

WORKING :

When some amount of heat produces electron gun starts liberating electron in which Thermionic Emission takes place. The electrons passes from the condenser lense or electromagneti lense in a particular width and falls on the sample. Then, the beams passed from condenser lense which amplifies the electrical signal and falls on the fluorescent screen by result of which it gives high quality images with high clarity, produce permanent images and 2D-image.

ADVANTAGES :

- It can produce permanent images.
- It is easy to train & use the TEM.

DISADVANTAGES :

- Very expensive to purchase.
- They're very big to handle.

Physics Q-Bank

5) Write a short note on thermal and radiation sensors?

Ans: (Refer Q no 4 & 5 of 5 marks)

6) Explain atomic force microscope?

Ans: (5 marks Q no 8)

* A He:Ne laser gives an output of ~~5mW~~ 5mW. The wavelength of the emitted radiation is $6328 \text{ \AA}$. Find the number of photons emitted per second.

Solⁿ Given:

$$P = 5 \text{ mW} = 5 \times 10^{-3} \text{ W}, \quad h = 6.63 \times 10^{-34}$$

$$\lambda = 6328 \text{ \AA} = 6328 \times 10^{-10} \text{ m}, \quad c = 3 \times 10^8 \text{ m/s}$$

$$N = \frac{PA}{hc}$$

$$= \frac{5 \times 10^{-3} \times 6328 \times 10^{-10}}{6.63 \times 10^{-34} \times 3 \times 10^8}$$

$$= \frac{31640 \times 10^{-13}}{19.89 \times 10^{-26}}$$

$$= 1590.7 \times 10^{13}$$

$$= 1.59 \times 10^{16} \text{ atoms}$$